

1. SECTION: PRODUCT AND COMPANY IDENTIFICATION

Product name : 伏泰
VORTEXX

Other means of identification : Not applicable

Recommended use : Sanitizer - Food contact surface

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.1 % - 2.2 %

Company : Ecolab Limited
10F., No.172, Sec.4, Chengde Rd., Shilin Dist., Taipei, Taiwan 111
(02) 8978-1234

Emergency telephone : 00801-14-8954 (Taiwan)
1-651-222-5352 (United States)

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2. SECTION: HAZARDS IDENTIFICATION
GHS Classification
Product AS SOLD

Organic peroxides : Type F
Acute toxicity (Oral) : Category 4
Acute toxicity (Inhalation) : Category 4
Acute toxicity (Dermal) : Category 5
Skin corrosion/irritation : Category 1B
Serious eye damage/eye irritation : Category 1
Short-term (acute) aquatic hazard : Category 3

Product AT USE DILUTION

Not a hazardous substance or mixture.

GHS label elements
Product AS SOLD

Hazard pictograms :



Signal Word : Danger

Hazard Statements : Heating may cause a fire.
Harmful if swallowed or if inhaled.
May be harmful in contact with skin.
Causes severe skin burns and eye damage.
Harmful to aquatic life.

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Precautionary Statements

: **Prevention:**

Keep away from heat/ sparks/ open flames/ hot surfaces. No smoking. Keep/Store away from clothing/ combustible materials. Keep only in original container. Avoid breathing mist or vapors. Wash skin thoroughly after handling. Do not eat, drink or smoke when using this product. Use only outdoors or in a well-ventilated area. Avoid release to the environment. Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

IF SWALLOWED: Call a POISON CENTER/ doctor if you feel unwell. Rinse mouth. IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor. Call a POISON CENTER/ doctor if you feel unwell. Wash contaminated clothing before reuse.

Storage:

Store locked up. Protect from sunlight. Store away from other materials.

Disposal:

Dispose of contents/ container to an approved waste disposal plant.

Product AT USE DILUTION

Precautionary Statements

: **Prevention:**

Wash hands thoroughly after handling.

Response:

Get medical advice/ attention if you feel unwell.

Storage:

Store in accordance with local regulations.

Product AS SOLD

Other hazards

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

3. SECTION: COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
Acetic acid	64-19-7	10 - 30
Hydrogen peroxide	7722-84-1	5 - 10
Peroxyacetic acid, type F, stabilized	79-21-0	1 - 5
Secondary Alkanesulphonates	5324-84-5	1 - 5
Octanoic acid	124-07-2	1 - 5

Product AT USE DILUTION

No hazardous ingredients

4. SECTION: FIRST AID MEASURES

First aid measures for different exposure routes

Product AS SOLD

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In case of eye contact	: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
In case of skin contact	: Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
If swallowed	: Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
If inhaled	: Remove to fresh air. Treat symptomatically. Get medical attention.
Protection of first-aiders	: If potential for exposure exists refer to Section 8 for specific personal protective equipment.
Notes to physician	: Treat symptomatically.
Most important symptoms and effects, both acute and delayed	: See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

5. SECTION: FIRE-FIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during fire fighting	: Special protective equipment for fire-fighters Oxidizer. Contact with other material may cause fire. Oxidizer; material is an oxidizer which may readily react with other materials, especially upon heating. Exposure to decomposition products may be a hazard to health.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides Sulfur oxides
Special protective equipment for fire-fighters	: In case of fire, wear a full face positive-pressure self contained breathing apparatus and protective suit.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local

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regulations. In the event of fire and/or explosion do not breathe fumes.

6. SECTION: ACCIDENTAL RELEASE MEASURES

Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Isolate the waste do not allow it to come into contact with incompatible materials. For small spills contain with sand or vermiculite and dilute the contained product at least 10 times with water. Transfer to an open topped container and remove to a safe place for neutralization* / disposal. For large spills contain spill and evacuate the area, leave until the reaction subsides, then collect up for disposal. Obtain consent from the local water company / authority if considering discharge to sewer. *NEUTRALIZATION : once diluted, neutralize with a suitable alkali such as sodium bicarbonate. Combustible materials exposed to this product should be rinsed immediately with large amounts of water to ensure that all product is removed. Residual product which is allowed to dry on organic materials such as rags, cloths, paper, fabrics, cotton, leather, wood, or other combustibles may spontaneously ignite and result in a fire.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.

Environmental precautions : No special environmental precautions required.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

7. SECTION: HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Do not ingest. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Do not mix with bleach or other chlorinated products – will cause chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep in a cool, well-ventilated place. Keep away from reducing agents. Keep away from combustible material. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers. Pressure bursts may occur due to gas evolution if the container is not adequately vented.

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Storage temperature : -30 °C to 40 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE). For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

8. SECTION: EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
Acetic acid	64-19-7	TWA	10 ppm 25 mg/m ³	TW OEL
		STEL	15 ppm 37.5 mg/m ³	TW OEL
Hydrogen peroxide	7722-84-1	TWA	1 ppm 1.4 mg/m ³	TW OEL
		STEL	2 ppm 2.8 mg/m ³	TW OEL

Biological occupational exposure limits : Contains no substances with biological exposure indices.

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

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Personal protective equipment

Eye protection	: No special protective equipment required.
Hand protection	: No special protective equipment required.
Skin protection	: No special protective equipment required.
Respiratory protection	: No personal respiratory protective equipment normally required.

9. SECTION: PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Color	: colorless	colorless
Odor	: pungent	vinegar-like
pH	: 0.9, (100 %)	2.3 - 4.1
Flash point	: Not applicable	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: No data available	
Evaporation rate	: No data available	
Flammability (solid, gas)	: Not applicable	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: 1.082	
Water solubility	: No data available	
Solubility in other solvents	: No data available	
Partition coefficient: n-octanol/water	: No data available	
Autoignition temperature	: No data available	
Thermal decomposition	: No data available	
Viscosity, kinematic	: No data available	
Explosive properties	: No data available	
Oxidizing properties	: No data available	
Molecular weight	: No data available	
VOC	: No data available	

10. SECTION: STABILITY AND REACTIVITY

Product AS SOLD

Reactivity	: No dangerous reaction known under conditions of normal use.
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Chemical stability	: pressure build-up Contamination may result in dangerous pressure increases - closed containers may rupture.
Possibility of hazardous reactions	: Do not mix with bleach or other chlorinated products – will cause chlorine gas.
Conditions to avoid	: Direct sources of heat. Exposure to sunlight.
Incompatible materials	: Bases Metals Organic materials
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides Sulfur oxides

11. SECTION: TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns. May be harmful in contact with skin.
Ingestion	: Harmful if swallowed. Causes digestive tract burns.
Inhalation	: May cause respiratory tract irritation. Harmful if inhaled. May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Pain, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

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Product AT USE DILUTION

Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

Product AS SOLD Product

Acute oral toxicity	: Acute toxicity estimate : 320 mg/kg Test substance: Product
Acute inhalation toxicity	: 4 h Acute toxicity estimate : 1.2 mg/l Test substance: Product Test atmosphere: dust/mist
Acute dermal toxicity	: Acute toxicity estimate : 3,706 mg/kg
Skin corrosion/irritation	: No data available
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: The substance or mixture is not classified as specific target organ toxicant, single exposure.
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

12. SECTION: ECOLOGICAL INFORMATION

Ecotoxicity

Product AS SOLD

Environmental Effects	: Harmful to aquatic life.
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Product AT USE DILUTION

Environmental Effects	: This product has no known ecotoxicological effects.
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Product AS SOLD Product

Toxicity to fish	: 96 h LC50 Pimephales promelas: 17.5 mg/l
Toxicity to daphnia and other aquatic invertebrates	: 48 h EC50 Daphnia magna (Water flea): 15.1 mg/l
Toxicity to algae	: No data available

Components

Toxicity to algae	: Acetic acid 72 h EC50 Skeletonema costatum (marine diatom): > 1,000 mg/l
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Hydrogen peroxide
72 h EC50: 1.38 mg/l

Peroxyacetic acid, type F, stabilized
72 h EC50: 0.7 mg/l

Persistence and degradability

Product AT USE DILUTION

Not applicable - Biocide

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

13. SECTION: DISPOSAL CONSIDERATIONS

Product AS SOLD

- Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

- Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

14. SECTION: TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

- UN number : 3109
- Proper shipping name : ORGANIC PEROXIDE TYPE F, LIQUID
(Peroxyacetic acid, type F, stabilized)
- Class : 5.2 (8)
- Environmentally hazardous : no

Air transport (IATA)

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Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

UN number : 3109
Proper shipping name : ORGANIC PEROXIDE TYPE F, LIQUID
(Peroxyacetic acid)
Class : 5.2 (8)
Marine pollutant : no
Special precautions for user : None

15. SECTION: REGULATORY INFORMATION

National regulatory information

Regulations on Labelling and Hazard Communication of Hazardous Chemicals

National regulatory information

Regulations on Occupational Safety and Health Facilities

National regulatory information

Standards for the Storage, Cleanup, Handling and Disposal of Industrial Waste

National regulatory information

Rules on Road Traffic Safety

National regulatory information

Standards of Permissible Exposure Limits in Job Site

National regulatory information

Regulations on Occupational Safety and Health Facilities

National regulatory information

Standards for the Storage, Cleanup, Handling and Disposal of Industrial Waste

National regulatory information

Regulations on Labelling and Hazard Communication of Hazardous Chemicals

National regulatory information

Rules on Road Traffic Safety

National regulatory information

Standards of Permissible Exposure Limits in Job Site

The ingredients of this product are reported in the following inventories:

Switzerland. New notified substances and declared preparations :
not determined

United States TSCA Inventory :
All substances listed as active on the TSCA inventory

Canadian Domestic Substances List (DSL) :
This product contains one or several components listed in the Canadian NDSL.

Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :

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On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemical Substances :
not determined

Japan. ENCS - Existing and New Chemical Substances Inventory :
On the inventory, or in compliance with the inventory

Korea. Korean Existing Chemicals Inventory (KECI) :
On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :
This product and/or component(s) are exempt or excluded from the Philippines Inventory of Chemicals and Chemical Substances (PICCS) under the Republic Act 6969 (RA 6969).

China. Inventory of Existing Chemical Substances in China (IECSC) :
On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory (TCSI) :
not determined

16. SECTION: OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet
Globally Harmonized System of Classification and Labeling of Chemicals (GHS)
IARC: (International Agency for Research on Cancer)
US. National Toxicology Program (NTP) Report on Carcinogens
ECHA List of Publishable Substances Registered
EU HPVCs (High Production Volume Chemicals)

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REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.